

Condition-Based Maintenance for Queues With Degrading Servers

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Abstract—The integration of condition monitoring with queueing systems to support decision making is not well explored. This paper addresses the impact of condition monitoring of the server on the system-level performance experienced by entities in a queueing system. The system consists of a queue with a single-server subject to Markovian degradation. The model assumes a Poisson arrival process with service times and repair times according to general distributions. We develop stability conditions and perform steady-state analysis to obtain performance measures (average queue length, average degradation, and so on). We propose minimizing an objective function involving four types of costs: repair, catastrophic failure, quality, and holding. The queue performance measures derived from steady-state analysis are benchmarked and compared to those from a discrete event simulation model. After verifying the queueing model, a sensitivity analysis is performed to determine the relationships between system performance and model parameters. Results indicate that the total cost function is convex and, thus, subject to an optimal repair policy. The model is sensitive to service time, quality costs, and failure costs for late-stage policy repairs decisions and sensitive to expected repair times and repair costs for early stage policy repair decisions.

Note to Practitioners—We propose condition-based maintenance for queues with degrading servers. The proposed model has been verified in simulation using degradation data of tools in a manufacturing facility. In order to implement, one must first model the underlying degradation process using the data from condition monitoring. We can get service and repair time distributions using data from the plant floor. Then, we need reasonable estimates of costs. These include costs of holding, quality, repair, and failure. They can be relative, prioritized or zero. The long-run average cost per unit time can then be evaluated. Then, the model provides the degradation state of the server at which maintenance is the most cost efficient. The degradation data can be used to continually obtain the optimal state of degradation at which the server can be taken offline with minimal costs.

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I. INTRODUCTION

SERVER degradation is an unavoidable part of service—and product-based industries. This paper focuses on a manufacturing setting where machines are considered as servers that can degrade over time. These servers often experience different forms of operational degradation ranging from tool wear to inefficient machine operation. Server degradation has a profound effect on product quality and can significantly impact bottom line profits of a company. As a result, efficient maintenance strategies are required to optimize repair schedules. Frequent maintenance can result in excessive repair and part replacement, which leads to increased machine unavailability. In contrast, delayed maintenance increases the risk of unexpected failure leading to higher corrective maintenance costs and excessive downtime. Wireman [1] reported that the maintenance cost in US industries increased by 10%–15% per year since 1979. The benchmarking study conducted by Wireman [1] estimated the maintenance costs for a selected group of industries to be 600 billion dollars in 1989. The operations and maintenance costs are 10%–20% of the total cost of energy (3.5–6 cents/kWh) for a wind farm [2]. While maintenance has a rich research legacy, condition-based maintenance where degradation data are used to optimize maintenance decisions is still an active and emerging area of research (see [3] and [4] for comprehensive reviews). Jardine *et al.* [5] provide an extensive review of various aspects of condition-based maintenance and their benefits. The majority of the existing literature in condition-based maintenance and degradation modeling has primarily focused on statistical data-driven approaches for quantifying degradation states and predicting remaining lifetime [6]–[8]. References [9] and [10] provide more specific details of costing models when it comes to using condition monitoring technologies and the cost-benefit tradeoff. Apart from fault detection, diagnostics, and prognostics, very little emphasis has been placed on the impact of degradation on the performance and operation of the system as a whole, i.e., the effects of machine degradation on the dynamics of in-process inventories, production delays, and system stability measures.

This paper impacts numerous manufacturing settings, both discrete and continuous, that involve capital-intensive machining servers whose degradation can result in significant economic losses. The pulp and paper industry is a classic example

of a multimillion dollar single-server facility. Paper mills usually consist of one, maybe two, paper machines. The raw material is fed through one end of a paper machine and finished products are output from the opposite end, i.e., a classic case of process manufacturing involving a single server. Degradation in this setting can result in a major shutdown that averages in the millions of dollars due to the massive amount of processed material that gets scrapped and the large amount of lost production. Other examples range from food and pharmaceutical to semiconductor industries.

This paper adopts a queueing theory perspective to machine degradation. Machining server degradation is modeled using the data acquired through condition monitoring, the process of using sensors to monitor a system's state of health. The costs associated with the system's performance/operation are assessed using queueing factors, loss in part quality, and other traditional factors related to repair costs and catastrophic failure. The objective is to develop a strategy that determines when to perform server maintenance on a single-server queue. Specifically, we are interested in determining the optimal degradation state at which the server should be taken off line. We derive an objective cost function consisting of repair, catastrophic failure, quality, and inventory holding costs. Our goal is to minimize the long-run average cost.

The dynamics of jobs arriving to the machine, getting processed one by one and leaving can be modeled as a queueing system with a single server. Hence, it is natural to consider queueing theory. However, the literature in queueing models only consider servers toggling between being up and down for random times [11], [12]. This paper extends those results in queueing theory to the case where machine or tool states can be monitored. Thus, the server instead of being in two states (up and down) can now be modeled as being in a variety of states. The main contribution to queueing theory is the extension of the well-studied M/G/1 queue to incorporate multiple degradation states of the server. In addition, we provide stability condition for such queues. Furthermore, we provide cost metrics, which are the time averages, by using departure-point probabilities. These are the major contributions to the wider field of queueing theory.

The rest of this paper is organized as follows. Section II presents a literature review, focusing on condition-based maintenance and queues with interruptions and vacations. Section III introduces the queueing model, develops transition probabilities, defines the objective function, derives queue stability conditions, and provides a step-by-step procedure for modeling the degradation process. Section IV presents the steady-state analysis of the queue followed by a simulation-based model verification and sensitivity analysis of model parameters. Finally, Section V gives the concluding remarks along with suggestions for future extensions of this paper.

II. LITERATURE REVIEW

The literature on the queueing side is of two types, namely, queues with interruptions and queues with server vacation. The division is based on the time the server is taken off.

Queues with interruptions deal with the server being taken off before the completion of the service. Such interruptions can be initiated by the customer or because of an emergency break the server requires. If the service is stopped upon completion of service, it is known as a vacation. A server may take a vacation if it needs a repair or if the queue is empty. Krishnamoorthy *et al.* [11] presented a survey on queues with interruptions. The survey covers both continuous and discrete time queueing models. They further subdivide each category based on the nature of the service times, number of servers, and so on. They also identify and summarize the tools for analyzing queueing models such as Z transform, the embedded Markov chain, the supplementary variables, and the matrix-analytic methods. Doshi [12] provides a survey on queues with vacation. He studies the models where the server takes vacation whenever it is idle. Doshi [12] discusses in detail two models (M/G/1 and G/G/1) and shows how these models can be related to other variants of queues with vacation.

Maintenance planning is a key to sustainable manufacturing, with various emerging technologies playing a significant role [13]. Researchers have developed many maintenance strategies with a range of objectives. Jirutitijaroen [14] study maintenance and failure costs of a degrading transformer as a function of the inspection rate of each state, mean time in each state, and transition probabilities from one state to another. They base the decision to perform maintenance on the comparison of maintenance and failure costs. Kaihara *et al.* [15] use Lagrangian decomposition coordination to minimize overall tardiness of a reentrant production facility. They assume maintenance is an additional job and schedule maintenance and production simultaneously achieving zero tardiness. Iung *et al.* [16] extend a probabilistic and event-based prognosis process to include a choice of maintenance strategies for a degrading system. The resulting model helps to select the best maintenance strategies to achieve improved system performance. The most efficient maintenance alternative is the one that has the minimum direct and indirect costs. Some researchers are working on combining production, quality, and maintenance. Bouslah *et al.* [17] optimize integrated production, quality control, and maintenance under the constraint of average outgoing quality limit. They give a simulation-based solution to the optimization problem and minimize the total incurred cost to find optimal production lot size, inventory threshold, sampling plan parameters, and overhaul threshold. Furthermore, they use type-1 continuous sampling plan for continuous-flow production systems with the same objective [18]. Similarly, Colledani and Tolio [19] study a multistage deteriorating manufacturing system and integrate production, quality, and maintenance control. They estimate the output effective production rate of the system to decide the critical state in which preventive maintenance should be activated. Wan *et al.* [20] propose active preventive maintenance using manufacturing big data. Ko and Byon [21] work on a large-scale system of homogeneous units subject to a non-Markovian degradation. They find the optimal thresholds for the fraction of units in each degradation state to trigger maintenance by minimizing the maintenance costs.

Each maintenance involves renewal of all units in the system simultaneously. Chan and Asgarpour [22] present an optimum maintenance policy for a component subject to random failures and failures due to deterioration. The problem is set as a Markov decision process and aims at maximizing the availability of the component to obtain the optimal mean time to preventive maintenance. Jin [23] studies a deteriorating system with a range of operations in a variable environment. Using information about system environment and deterioration, the author develops a 2-D control limit policy that selects operations with minimum long-run expected cost. Byon and Ding [24] use backward dynamic programming to solve a partially observed Markov decision process model for wind turbines. Their objective is to minimize operation and maintenance costs to choose between preventive maintenance, on-site observation, and no action. Xanthopoulos *et al.* [25] also integrate production/maintenance control for a deteriorating, stochastic system but propose reinforcement learning for deriving optimal policies.

Dvorak [26] highlights the differences between condition-based and interval-based oil changes in wind turbines. While the diagnostics and laboratory analysis is expensive and location and weather variable, Dvorak [26] expects that condition-based maintenance might be more cost efficient and safer for the future of wind energy. Elwany and Gebraeel [7] incorporate remaining life distributions (RLDs) in an existing replacement and inventory decision model [27]. They obtain these distributions using sensory data obtained from condition monitoring of deteriorating components. When compared to those obtained using population failure time distributions, RLDs give a more accurate prediction of optimal times for component replacement and spare parts ordering times. Elwany *et al.* [8] use real-time condition monitoring to model the underlying degradation process of a deteriorating component. They obtain an optimal replacement policy through a Markov decision process considering costs of replacement, failure, and observation. Tian and Lio [6] study condition-based maintenance for a multicomponent system. The components are assumed to be identical while exhibiting economical dependencies. A Proportional Hazards Model is the basis of their optimal condition-based maintenance policy. The cost is a function of risk threshold values and consists of repair and failure costs. Castanier *et al.* [28] consider condition-based maintenance for a two-unit system. Their policy consists of component- and system-level maintenance and the determination of next system inspection time. They use a cost model including costs of inspection, corrective and preventive maintenance, and failure to determine repair thresholds for two-unit system. Maillart [29] examines the problem of adaptively scheduling observations with condition-based maintenance. The author models the system as a partially observable Markov decision model with the objective of minimizing costs of observation, maintenance, and failure. Shafiee and Finkelstein [30] study condition-based group maintenance for a system with nonidentical subsystems. They look for the alert threshold values for subsystems and the replacement age for a whole system by minimizing the cost. Wang *et al.* [31] introduce scheduling and

maintenance thresholds in a framework of condition-based maintenance.

None of the work in the literature addresses degrading servers and maintenance decisions from both queueing and quality standpoints. Clearly, delays due to server maintenance can have significant impact on time in system, average queue length, system stability, and so forth, and a poorly maintained server condition can produce products of substandard quality. Thus, research addressing condition-based maintenance from the queueing and quality perspectives is very relevant.

In this paper, we model a single-server queue with Markovian server degradation. The server condition is observed at every service completion and a decision is made on whether or not to perform maintenance. We use a multidimensional cost function which includes costs for waiting (or holding), quality, usual maintenance, and catastrophic failure. Our goal is to identify a decision policy which keeps the system stable and minimizes total system cost over the long term.

III. MODEL FORMULATION AND ANALYSIS

Consider a single-stage queueing system with a single waiting line and a single server. Entities arrive one by one, get served according to a first-come-first-served discipline, and leave. The server degrades over time as entities are processed. Entities arrive into the queueing system according to a Poisson process with rate λ arrivals per hour. The service times (represented by a generic random variable S) are independent and identically distributed with common cumulative distribution function (CDF) $G(t) = P\{S \leq t\}$. There is an infinite waiting area. At the end of every entity's service, the condition of the server is measured as one of $M+1$ states. The states $\{0, 1, 2, \dots, M-1\}$ are considered operational and state M denotes a catastrophic failure. As the server's condition degrades, the measured value becomes higher. Let Y_n be the server's condition immediately after the n th service completion. Let d_{ij} be the probability that the $n+1^{st}$ service completion results in an observed value of j given that the measurement at the n th completion was i , that is,

$$d_{ij} = P\{Y_{n+1} = j | Y_n = i\} \quad (1)$$

for all i and j in $\{0, 1, \dots, M\}$. We assume that d_{ij} is either given or can be computed.

We consider the following strategy: At the end of a service, if the server condition is measured as m or higher, *corrective action* such as replacing a tool (or the server itself) shall be performed. The time to perform the corrective action is assumed to be generally distributed (represented by a random variable R) with CDF $F(t) = P\{R \leq t\}$. We assume that once the corrective action is complete, the server starts in degradation state 0. For the time being, it is assumed that a catastrophic failure also takes the same time R to take corrective action (this can be relaxed). However, a much larger replacement cost is incurred. We assume that arrivals continue to occur when corrective action is performed but no service activity occurs. It is assumed that entities do not renege (and for that matter, they do not balk upon arrival). Our goal is to optimally select a degradation state m such that the long-run average cost per unit time is minimized.

In summary, the model assumes a Poisson arrival process with service times and repair times according to general distributions. The ongoing service gets completed before the server is taken off for repair. Arrivals continue to occur when corrective action is performed but no service activity occurs. It is assumed that entities do not renege (and for that matter, they do not balk upon arrival). Once the corrective action is complete, the server starts in degradation state 0. The server, which starts in degradation state 0, fails in degradation state M . Catastrophic failure also takes the same time R to take corrective action (this can be relaxed). However, a much larger replacement cost is incurred. We assume that d_{ij} is independent of service time distribution. Quality cost increases with server degradation.

Let X_n be the number of entities in the system immediately after the n th departure. Let $\hat{Y}_n = \min(Y_n, m)$ by collapsing observations $m, \dots, M$ to a single state m . We can model the bivariate stochastic process $\{(X_n, \hat{Y}_n), n \geq 0\}$ as a Markov chain on state space given by the cartesian product $\{0, 1, 2, \dots\} \times \{0, 1, \dots, m\}$. The transition probability matrix P of the Markov chain has the following form:

$$P = \begin{bmatrix} B_0 & B_1 & B_2 & B_3 & \dots \\ A_0 & A_1 & A_2 & A_3 & \dots \\ & A_0 & A_1 & A_2 & \dots \\ & & A_0 & A_1 & \dots \\ & & & \ddots & \ddots & \ddots \end{bmatrix} \quad (2)$$

where A_k and B_k , $k \geq 0$ are $(m+1) \times (m+1)$ matrices. The transition probability matrix P is of the generic M/G/1-type that has been studied by Akar *et al.* [32]. The submatrices A_k and B_k have been specifically derived in our paper.

The P matrix in (2) is the transition probability matrix corresponding to the bivariate stochastic process that keeps track of the number of jobs in the system and the condition of the server, both immediately after a job's processing is completed (i.e., end of service). There are some special cases to be careful about: 1) if at the end of service, the system becomes empty; 2) if at the end of service, the machine tool is replaced; or 3) both cases (i) and (ii). Thus, the ij th element of matrix A_k is written as $A_k(i, j)$ and defined as the probability that given the server is observed in state i at the end of a service, the server would be observed in state j at the end of the next service, and k new jobs arrive into the system between the two service completions. Note from (2) that the states where A_k is used are when there is at least one job in the system when observed after a service completion (i.e., when the machine deterioration state is i). Hence, the time between consecutive services can either be S (the service time) or $S + R$ a time to repair plus time to serve. They correspond to $i < m$ and $i = m$, respectively, because in the case $i < m$, there would be no replacement and time between observations is just a service. However, when $i = m$, the server is replaced, taking R time and then the service is started and lasts for S time. Hence, we get (4). The k term is the standard upper-Hessenberg matrix structure found in standard M/G/1 queues. In fact, the special case of

the server not deteriorating would result in A_k being a scalar and identical to that in an M/G/1 queue's transition probability.

The B_k matrices are a little more involved. Note that they show up only on the first row of P in (2). They correspond to the cases when at the end of a service, the system becomes empty. We consider the case $i < m$ first where the machine tool is not replaced. The first event is a new job arrival. This would be the job whose departure would correspond to the next observation. Thus, we need to only consider the number of arrivals k during this service time S , which would result in k jobs in the system at that observation. Now, let us consider the case $i = m$. Here, the machine tool is replaced which takes R units of time. During this time R , either no jobs arrive (in which case we wait for the next job to arrive and during its service time S we obtain the probability of k new arrivals). However, if at least one arrival occurs during the tool replacement time R , then the probability that k jobs arrive after the first one (i.e., the one that would depart at the next observation) which happens after time U would need to be computed in time $S + R - U$. In other words, we are interested in S if $R < U$, and in $S + R - U$ if $R > U$. Hence, we write it as the number of arrivals in time $S + (R - U)^+$ which is $S + \max(R - U, 0)$. This results in (5). The special case where a server does not deteriorate, ends up in scalar values of $B_k = a_k$ for all k . Again, this would correspond to the first row of the P matrix of the standard M/G/1 queue which would be identical to the second row.

To obtain the entities of A_k and B_k , consider the following: For all $k \geq 0$, let a_k , b_k , and c_k be the probabilities that k arrivals occur during time S , time $S + R$, and time $S + (R - U)^+$, respectively, where $U \sim \exp(\lambda)$ corresponds to an interarrival time. For an arbitrary random variable Z with CDF $H(t) = P\{Z \leq t\}$ since the arrivals are Poisson with rate λ , we can compute the probability that k arrivals occur during time Z as

$$\int_0^\infty e^{-\lambda t} \frac{(\lambda t)^k}{k!} dH(t) \quad (3)$$

for all $k \geq 0$. Now, we can write down the (i, j) th entry of A_k and B_k for any $i \in \{0, 1, \dots, m\}$ and $j \in \{0, 1, \dots, m\}$ as follows:

$$A_k(i, j) = \begin{cases} a_k d_{ij} & \text{if } i < m \text{ and } j < m \\ a_k \left(1 - \sum_{\ell=0}^{m-1} d_{i\ell}\right) & \text{if } i < m \text{ and } j = m \\ b_k d_{0j} & \text{if } i = m \text{ and } j < m \\ b_k \left(1 - \sum_{\ell=0}^{m-1} d_{0\ell}\right) & \text{if } i = m \text{ and } j = m \end{cases} \quad (4)$$

$$B_k(i, j) = \begin{cases} a_k d_{ij} & \text{if } i < m \text{ and } j < m \\ a_k \left(1 - \sum_{\ell=0}^{m-1} d_{i\ell}\right) & \text{if } i < m \text{ and } j = m \\ c_k d_{0j} & \text{if } i = m \text{ and } j < m \\ c_k \left(1 - \sum_{\ell=0}^{m-1} d_{0\ell}\right) & \text{if } i = m \text{ and } j = m \end{cases} \quad (5)$$

for all $k \geq 0$.

The detailed computations of a_k , b_k , and c_k for two cases with different repair time distributions are now presented.

A. Case 1: Constant Repair, Constant Service

Case 1 assumes constant repair times r and constant service times s , thus $R = r$ and $S = s$ with probability 1. Interarrival time is exponentially distributed with rate λ . Remember that a_k , b_k , and c_k are the probabilities that k arrivals occur during time S , time $S + R$, and time $S + (R - U)^+$, respectively, where $U \sim \exp(\lambda)$ corresponds to an interarrival time. The relation for a_k and b_k is straightforward using (3), and c_k is derived in the Appendix

$$a_k = e^{-\lambda s} \frac{(\lambda s)^k}{k!} \quad (6a)$$

$$b_k = e^{-\lambda(s+r)} \frac{(\lambda(s+r))^k}{k!} \quad (6b)$$

$$c_k = \frac{e^{-\lambda(s+r)} \lambda^k}{k!} \left(\frac{\lambda}{k+1} X((s+r)^{k+1} - s^{k+1}) + s^k \right). \quad (6c)$$

B. Case 2: Weibull Repair, Constant Service

For case 2, the repair time R follows a Weibull distribution with α is the shape parameter and β is the rate parameter, and service time $S = s$, a constant. We have Poisson arrivals with rate λ . The derivation for a_k and b_k is straightforward using (3), and c_k is derived in the Appendix

$$a_k = e^{-\lambda s} \frac{(\lambda s)^k}{k!} \quad (7a)$$

$$b_k = \int_0^\infty e^{-\lambda(s+r)} \frac{(\lambda(s+r))^k}{k!} \alpha \beta (r\beta)^{\alpha-1} e^{-(r\beta)^\alpha} dr \quad (7b)$$

$$c_k = \int_0^\infty e^{-\lambda s} e^{-(r\beta)^\alpha - \lambda r} \left((s+r)^{k+1} \frac{\lambda^{k+1}}{(k+1)!} \alpha \beta (r\beta)^{\alpha-1} \right) dr \\ - \int_0^\infty e^{-\lambda s} e^{-(r\beta)^\alpha - \lambda r} (s^{k+1}) \frac{\lambda^{k+1}}{(k+1)!} \alpha \beta (r\beta)^{\alpha-1} dr \\ + \int_0^\infty e^{-\lambda s} e^{-(r\beta)^\alpha - \lambda r} \frac{(\lambda s)^k}{k!} \alpha \beta (r\beta)^{\alpha-1} dr. \quad (7c)$$

Using (6) for Case 1 and (7) for Case 2, we can compute the (i, j) th entry of A_k and B_k for any $i \in \{0, 1, \dots, m\}$ and $j \in \{0, 1, \dots, m\}$ as defined in (4) and (5), respectively. The computation of the transition probability matrix P then follows as in (2).

C. Degradation Process

In order to build the transition probability matrix (P) of the Markov chain, a model for the degradation of the server is needed. Recall that d_{ij} from (1) is the probability that the $n+1$ th service completion results in an observed degradation state j , given that the measurement at the n th service completion was degradation state i . For now, assume that d_{ij} is independent of service time distribution. Under this assumption, we illustrate a few models for d_{ij} .

- 1) *Geometric Distribution*: For all i and j such that $0 \leq i \leq j < M$

$$d_{ij} = (1 - p_i)(p_i)^{j-i}$$

where p_i is a probability. In this case, $d_{ij} = 0$ for $j < i$ and d_{iM} is computed such that the rows of the matrix sum to 1 where

$$d_{iM} = 1 - \left(\sum_{j=0}^{M-1} d_{ij} \right). \quad (8)$$

- 2) *Poisson Distribution*: For all i and j such that $0 \leq i \leq j < M$

$$d_{ij} = e^{-ht} \frac{(ht)^{j-i}}{(j-i)!}$$

where $h = (1/\text{MTTF})$, MTTF is the mean time to failure and $t = E[S]$ or the mean service time. Again, $d_{ij} = 0$ for $j < i$ and d_{iM} is computed such that the rows of the matrix sum to 1 as in (8).

- 3) *Empirical Distribution Method*: Data from a deteriorating component in a manufacturing facility can be used to obtain average rates of transition from one state to another. We now present a method to deduce d_{ij} .

Suppose a new tool is monitored to collect degradation data until it fails. The process is repeated for V observations where each observation begins with the same tool “as good as new” until it fails. In order to obtain d_{ij} , the steps are as follows.

- 1) Set the catastrophic failure state M , where $M > 0$.
- 2) Normalize the data over all degradation states.
- 3) Select the breakpoints in the data representing transition from one degradation state to another.
- 4) If the data presents irregularities such as reverting to an earlier degradation state, consider defining a rule for removing these irregularities. Two potential rules are as follows.
 - a) The server is assumed to enter state $i+1$ when it has entered state $i+1$ for the n th time.
 - b) The server is assumed to enter state $i+1$ when it has left state i for n number of times.
- 5) Find the time T_{iv} spent by the tool in state i on observation v for all $i \in [0, M]$ and $v \in [1, V]$. This can be calculated by taking the difference of the first time at which the server was in state $i+1$ and the first time it was in the same state i .
- 6) Calculate the average time spent T_i in state i using

$$T_i = \left(\sum_{v=1}^V T_{iv} \right) / V.$$

- 7) Find rate of transition γ from state i to $i+1$ by taking inverse of T_i for all $M+1$ states

$$\gamma_i = 1/T_i.$$

- 8) Generate the infinitesimal generator matrix Q_d as follows:

$$Q_d = \begin{bmatrix} -\gamma_0 & \gamma_0 & 0 & 0 & \dots & 0 & 0 \\ 0 & -\gamma_1 & \gamma_1 & 0 & \dots & 0 & 0 \\ 0 & 0 & -\gamma_2 & \gamma_2 & \dots & 0 & 0 \\ 0 & 0 & 0 & -\gamma_3 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & -\gamma_M & \gamma_M \\ 0 & 0 & 0 & 0 & \dots & 0 & 0 \end{bmatrix}.$$

- 9) Take exponential of the matrix obtained by the product of the infinitesimal generator matrix Q_d and mean service time $E[S]$ to get d_{ij} for all i and j . The d_{ij} 's constitute the degradation matrix d

$$\begin{aligned} d &= \exp(Q_d \times E[S]) \\ &= I + \sum_{i=1}^{\infty} (Q_d \times E[S])^i / i. \end{aligned}$$

D. Stability Conditions

Next, we derive the stability condition for the queue under consideration. Since the single sever is degrading, stability conditions are dependent on the “mean time to repair” (ω), as shown in Theorem 1. Note that ω as defined here depends on repair state, m .

Theorem 1: Consider a single-server queue with degrading server, Poisson arrivals with rate λ , service and server repair times generally distributed with means $E[S]$ and $E[R]$, respectively, and mean time to server repair, ω . Then, the queue is stable when $\lambda < ((\omega/\omega + E[R]))(1/E[S])$.

Proof: Let h_i be the expected value of the number of parts completed from now until maintenance in state m given that the current degradation is level i . Then, $h_m = 0$ and $h_i = 1 + \sum_{j=i}^m h_j d_{ij}$ for $0 \leq i \leq m-1$, and the recursion becomes

$$\begin{aligned} h_m &= 0 \\ h_{m-1} &= \frac{1 + h_m d_{m-1,m}}{1 - d_{m-1,m-1}} = \frac{1}{1 - d_{m-1,m-1}} \\ &\dots \\ h_i &= \frac{1 + \sum_{j=i+1}^m h_j d_{ij}}{1 - d_{i,i}} \\ &\dots \\ h_0 &= \frac{1 + \sum_{j=1}^m h_j d_{0j}}{1 - d_{0,0}} \end{aligned}$$

where h_0 is the expected number of parts to be completed from state 0 to state m . Then, the mean time to repair (ω) is given by $\omega = h_0 E[S]$. The traffic intensity $\hat{\rho} = \lambda E[S](\omega + E[R]/\omega)$ and the queue are stable for $\hat{\rho} < 1$. Thus, the stability condition is $\lambda < (\omega/\omega + E[R])(1/E[S])$. ■

Assume that the Markov chain $\{(X_n, \hat{Y}_n), n \geq 0\}$ is stable. We can compute the stationary probability vector $\pi = [\pi_0 \pi_1 \pi_2 \dots]$ which satisfies

$$\pi = \pi P, \quad \pi \bar{1} = 1$$

where π_k for all $k \geq 0$ is $1 \times (1 + m)$ and $\bar{1}$ is an infinite column vector of ones.

E. Average Cost per Unit Time

We consider four aspects to determine the long-run average cost per unit time: 1) quality cost: if the state of the server is i when a product is completed (or a service is rendered), we consider a cost $C_Q q_i$ incurred where C_Q is a constant and q_i is a function that increases with i (i.e., more degradation implies more cost to the system); 2) holding cost: a cost C_H is

TABLE I
NUMERICAL VALUES FOR MODEL PARAMETERS

Parameter	Value
$E[S]$	3 seconds
$E[R]$	8 seconds
λ	0.267 per second
$\hat{\rho}$	≈ 0.8
M	6
K	30

incurred per unit time a product spends in the system; 3) repair cost: a cost C_R is incurred every time a server is repaired; 4) failure cost: a cost C_F is incurred every time a catastrophic failure occurs (i.e., the degradation state upon departure is M). Using $\pi_k(i)$ (the i th element of vector π_k) for all $k \geq 0$ and $i \in \{0, 1, \dots, m\}$, the long-run average cost per unit time $C(m)$ is

$$\begin{aligned} C(m) &= \sum_{k=0}^{\infty} \sum_{i=0}^m [\lambda C_Q q_i \pi_k(i) + \lambda C_F d_{rem(i,m)M} \pi_k(i) \\ &\quad + k C_H \pi_k(i) + \lambda C_R (i - rem(i, m)) \pi_k(i) / m] \end{aligned}$$

where q_i is the quality of product (or service) after which the server degeneration status is measured as i , and $rem(i, m)$ is the remainder when i is divided by m . The holding cost term $k C_H \pi_k(i)$ is based on Poisson arrivals see time averages and that departure point probabilities equal arrival point probabilities. Note that the departure rate is indeed λ .

IV. NUMERICAL ANALYSIS

In this section, we first discuss steady-state analysis of the model, including values for the model parameters. Next, we verify the model by comparing it to a simulation, then perform a sensitivity analysis on several model parameters. Verification and analysis are conducted using the data for the empirical distribution method outlined in Section III. Finally, the nature of the cost function is also discussed.

A. Steady State Analysis and Cost Function Convexity

The model was coded in MATLAB. Table I summarizes the model parameters. The expected service time was 3 s, the expected repair time was 8 s, and the arrival rate was 0.267 per second. Therefore, the server remains busy around 80% of the time (e.g., the traffic intensity $\hat{\rho} = 0.80$). The server, which starts in degradation state 0, was assumed to fail in degradation state $M = 6$. To numerically solve the steady-state distribution of the Markov chain, we truncate the queue length as a fairly large K .

To populate the degradation process for the server, vibration-based empirical data were collected for a new tool, which was condition monitored every 2 s until failure. The experiment was conducted for 20 new tools under similar testing conditions. Fig. 1 shows the degradation data collected from these observations and shows the evolution of vibration signal with time for each tool until it runs to failure. Gebrael *et al.* [33] presents a detailed understanding of such a degradation data.

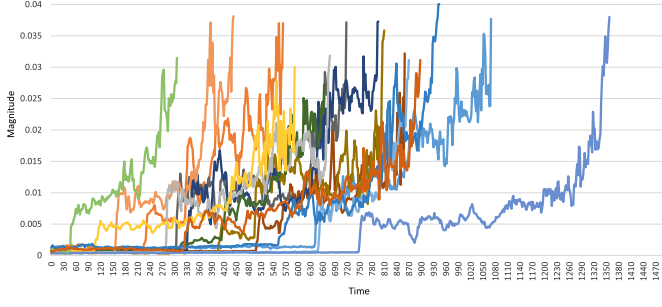


Fig. 1. Vibration data of a degrading tool.

TABLE II
RATES OF TRANSITION

Degradation State $i - i+1$	Rate (per second)
0-1	0.0029
1-2	0.0107
2-3	0.0094
3-4	0.0182
4-5	0.0152
5-6	0.0437

TABLE III
COST COEFFICIENTS

Coefficient	Value (cost units)
C_H	10
C_R	100
C_F	500
C_Q	50

Using the empirical distribution method with catastrophic failure state $M = 6$, the following rates of transition were obtained in Table II from step 7 of the empirical distribution method.

Based on the rates of transition from Table II and the expected service time, $E[S] = 3$, from Table I, we found the degradation matrix, d , from step 9 of the empirical distribution method to yield the following matrix:

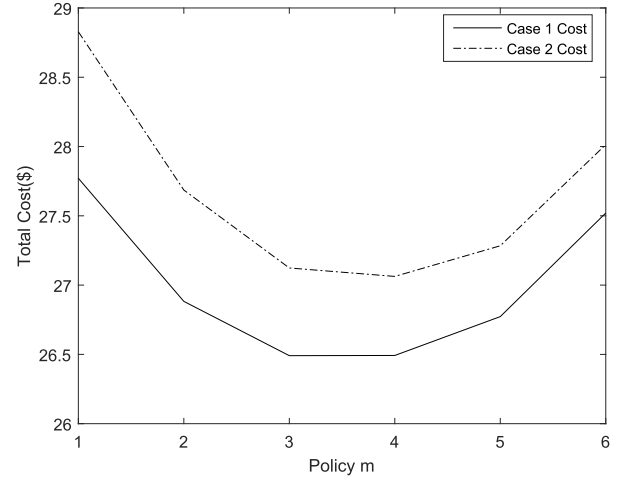
$$d = \begin{bmatrix} 0.9914 & 0.0085 & 0.0001 & 0 & 0 & 0 & 0 \\ 0 & 0.9684 & 0.0312 & 0.0004 & 0 & 0 & 0 \\ 0 & 0 & 0.9721 & 0.0271 & 0.0007 & 0 & 0 \\ 0 & 0 & 0 & 0.9470 & 0.0518 & 0.0011 & 0.0001 \\ 0 & 0 & 0 & 0 & 0.9556 & 0.0416 & 0.0028 \\ 0 & 0 & 0 & 0 & 0 & 0.8772 & 0.1228 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1.0000 \end{bmatrix}.$$

Next, Table III shows the cost coefficients. The choice of cost coefficients may vary from one application to another. Here, we choose the values such that $C_H < C_Q < C_R < C_F$. Naturally, $C_R < C_F$ so the failure cost is five times greater than repair cost. The holding and quality costs are lower, with quality costs being five times greater than the holding cost.

Finally, the Taguchi Loss Function [34] was used to determine the quality of the product (or service). The values were

TABLE IV
QUALITY

Degradation State	q_i
0	0.0029
1	0.0233
2	0.0787
3	0.1866
4	0.3644
5	0.6297
6	1.0000

Fig. 2. Cases 1 and 2: total cost against policy m .

normalized between 0 and 1, and the specific values used for each degradation state i are listed in Table IV.

Using the model parameters in Tables I–IV, steady-state probabilities were obtained and used to calculate average queue length, L , and the average degradation state, D . Finally, the cost function $C(m)$ was evaluated and plotted against m . Fig. 2 shows the results for both Cases 1 and 2. Note that the total cost, $C(m)$, when plotted against policy m , shows convex behavior. This implies that the objective is of convex optimization and an optimal repair policy, m , exists. The minimal cost in Fig. 2 for Case 1 occurs at $m = 3$ and for Case 2 at $m = 4$, which are the optimal repair states, respectively. For Case 1 at $m = 3$, $L = 2.605$, and $D = 0.574$, and for Case 2, at $m = 4$, $L = 2.642$, and $D = 0.800$.

Note that $C(m)$ cannot be written as a closed-form algebraic expression in terms of m to be able to determine its convexity. In particular, $\pi_k(i)$ is obtained by solving for the left eigenvector of P corresponding to the eigenvalue of 1. It is not possible to write down $\pi_k(i)$ in terms of m even for the simplest case of $m = 1$. However, it is possible to provide an intuitive explanation. The holding cost component decreases with m because the sum over all k and all i $k C_H \pi_k(i)$ is just $C_H L$ where L is the average number of jobs in the system. This is because the server is available a larger fraction of time as m increases L decreases. Since the server is available a larger fraction of time, there are fewer replacements per unit time, and hence, the repair costs reduce with increase in m . However, quality and failure costs increase with m .

TABLE V
CASE 1: COMPARISON BETWEEN NUMERICAL MODEL AND
SIMULATION FOR REPAIR POLICY STATE $m = 4$

Performance Measure	MATLAB	Simulation	Error
Total Cost	26.492	26.883	1.454%
Quality Cost	0.506	0.553	8.499%
Holding Cost	25.853	26.146	1.120%
Repair Cost	0.133	0.164	18.902%
Failure Cost	6.299×10^{-4}	0.019	96.684%
Avg. Queue Length (L)	2.585	2.615	1.147%
Avg. Degradation (D)	0.800	0.839	4.648%

TABLE VI
CASE 2: COMPARISON BETWEEN NUMERICAL MODEL AND
SIMULATION FOR REPAIR POLICY STATE $m = 4$

Performance Measure	MATLAB	Simulation	Error
Total Cost	27.063	27.491	1.556%
Quality Cost	0.506	0.561	9.803%
Holding Cost	26.423	26.737	1.174%
Repair Cost	0.133	0.162	17.901%
Failure Cost	6.299×10^{-4}	0.031	97.968%
Avg. Queue Length (L)	2.642	2.674	1.196%
Avg. Degradation (D)	0.800	0.842	4.988%

since a larger m would imply more products are made with deteriorated tool, and a higher chance of reaching M during a service. Thus as a combination, the total cost $C(m)$ exhibits a convex nature.

B. Model Verification

To further verify the results of our modeling analysis, we coded the queueing model and degrading server in a discrete event simulation using AnyLogic 7.3.6. Note that the analytical model gives an accurate result. Simulation, on the other hand, provides with a region of optimal solution. Moreover, computation time per run is shorter for the analytical model given the choice of model parameters. There are a few notable differences in how the simulation models operate versus the steady-state queueing model. For example, the bound on the queue size ($K = 30$) was not included in the simulation. In addition, the simulation model assumes that the server is degrading as time passes, regardless of whether the server is active (a reasonable assumption for human servers). In the Markov queueing model, server degradation only occurs during service and can only be observed after a service completion.

Monte Carlo simulation was used for 100 replications and the reported results are the average across all replications. Tables V and VI show the comparison of the results of the MATLAB steady-state analysis to the AnyLogic simulation model for Cases 1 and 2, respectively. In order to verify our MATLAB model, we compared the total cost, individual costs (quality, failure, repair, and holding), average queue length, and average degradation to those obtained from the discrete event simulation. The same policy repair state $m = 4$ was used in both the MATLAB model and the AnyLogic simulation. The total cost was found to have only a 1.454% error for Case 1 and 1.556% error for Case 2. The error for the average

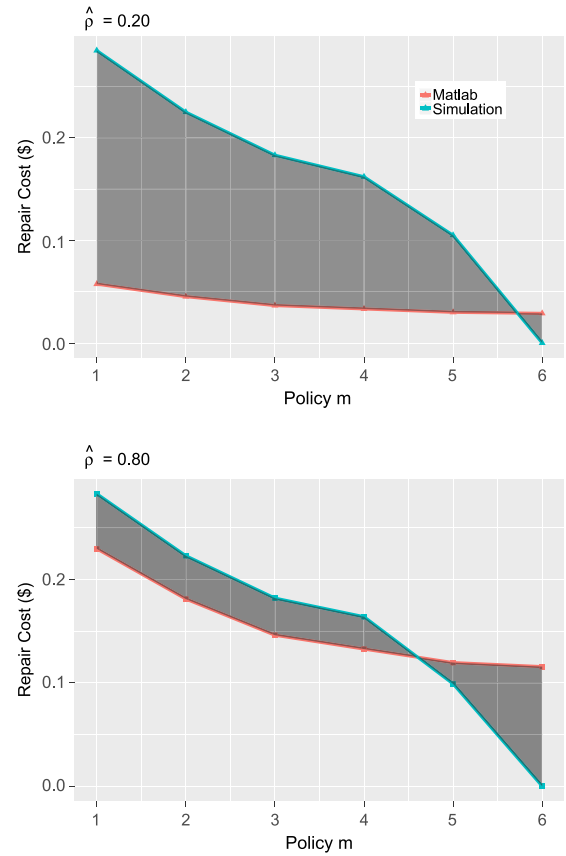


Fig. 3. MATLAB versus simulation for repair cost.

TABLE VII
CASE 1: COMPARISON BETWEEN NUMERICAL MODEL AND SIMULATION
FOR REPAIR POLICY STATE $m = 4$ (CONFIDENCE INTERVALS)

Performance Measure	MATLAB	Mean	Std. Dev.	99% CI
Total Cost	26.492	26.883	1.615	(26.467, 27.299)
Quality Cost	0.506	0.553	0.050	(0.540, 0.566)
Holding Cost	25.853	26.146	1.604	(25.733, 26.559)
Repair Cost	0.133	0.164	0.015	(0.160, 0.168)
Failure Cost	6.299×10^{-4}	0.019	0.018	(0.014, 0.024)
Avg. Queue Length (L)	2.585	2.615	0.160	(2.574, 2.656)
Avg. Degradation (D)	0.800	0.839	0.083	(0.818, 0.860)

queue length, L , was less than 2% and the error for the average degradation state, D , was less than 5%. Errors in the holding costs were less than 2% and the errors in the quality costs were 8%–10%. The largest discrepancies between the two models are in the repair costs (>18%) and failure costs (>97%). These large differences are due to the previously mentioned fact that the degradation process advances faster in the simulation model. We can further show evidence of this fact in Figs. 3 and 4. Using a small value of $\hat{\rho} = 0.20$, the server is not very busy and we can see a larger gap between the two models (i.e., larger shaded area) than for larger values of $\hat{\rho} = .80$ where the server is busier and, thus, degrading quickly like the simulation model. Therefore, the MATLAB and simulation models are suitable alike for higher traffic intensities.

Tables VII and VIII present 99% confidence intervals computed from the simulation study.

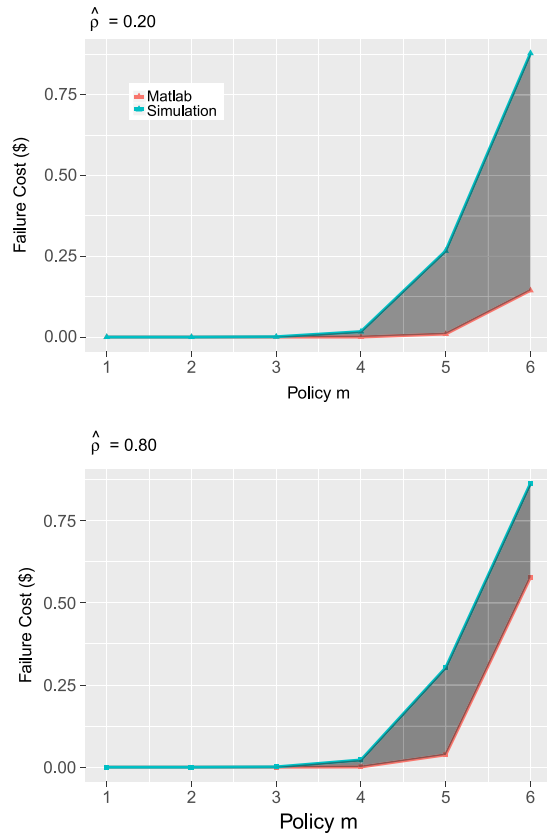


Fig. 4. MATLAB versus simulation for failure cost.

TABLE VIII
CASE 2: COMPARISON BETWEEN NUMERICAL MODEL AND
SIMULATION FOR REPAIR POLICY STATE $m = 4$
(CONFIDENCE INTERVALS)

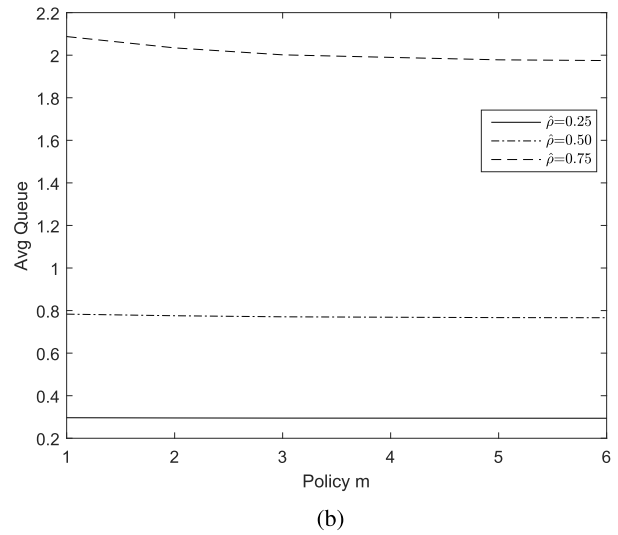
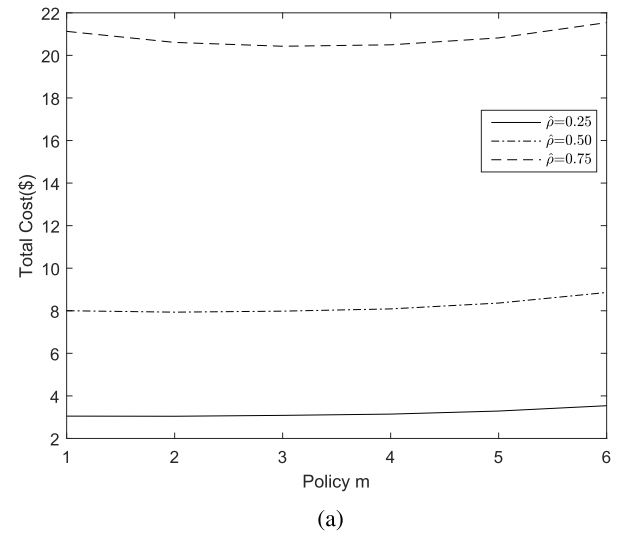
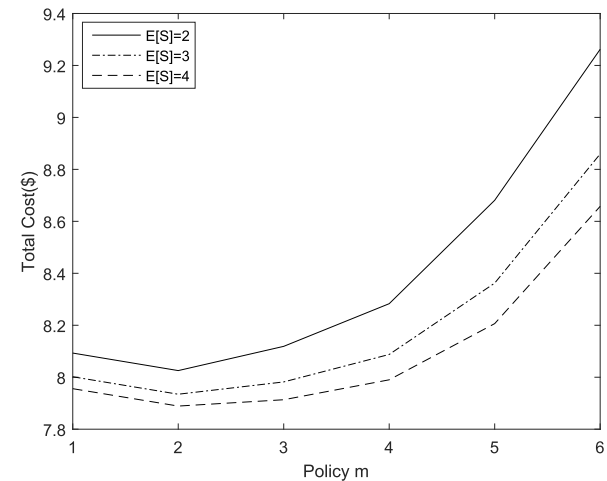
Performance Measure	MATLAB	Simulation		
		Mean	Std. Dev.	99% CI
Total Cost	27.063	27.491	2.237	(26.915, 28.067)
Quality Cost	0.506	0.561	0.067	(0.544, 0.578)
Holding Cost	26.423	26.737	2.215	(26.167, 27.308)
Repair Cost	0.133	0.162	0.014	(0.158, 0.166)
Failure Cost	6.299×10^{-4}	0.031	0.025	(0.025, 0.037)
Avg. Queue Length (L)	2.642	2.674	0.222	(2.617, 2.731)
Avg. Degradation (D)	0.800	0.842	0.089	(0.819, 0.865)

The total “MATLAB” cost was found to be within the 99% confidence interval for Case 1 and Case 2. The same was true for the average queue length, L , and the holding costs. On the other hand, the average degradation state, D , was found to be smaller than the lower bound of the 99% confidence interval. The quality costs being dependent on the average degradation state showed similar behavior. Again, the largest discrepancies between the two models are in the repair costs and failure costs, for reasons noted earlier.

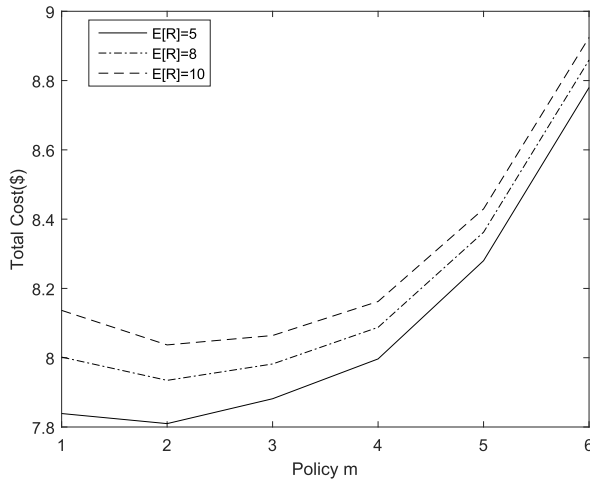
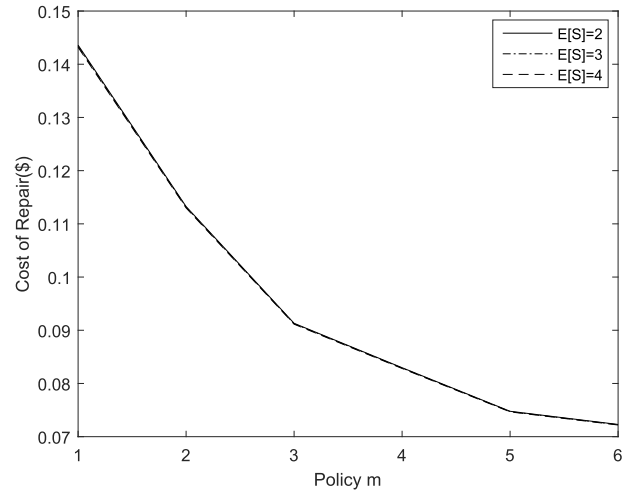
C. Sensitivity Analysis

A sensitivity analysis was conducted on several model parameters ($\hat{\rho}$, $E[S]$, and $E[R]$). Cases 1 and 2 exhibit similar trends, thus we present the results for Case 1 only (Figs. 5–7). Note that parameters are varied one at a time with initial values provided in Table I.

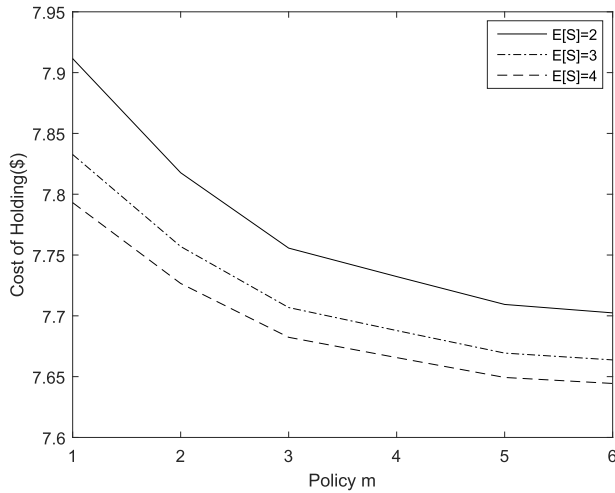
Observation 1: As traffic intensity increases, total cost and average queue length also increase (Fig. 5).

Fig. 5. Case 1: sensitivity analysis for $\hat{\rho}$. (a) Total cost. (b) Average queue.Fig. 6. Case 1: sensitivity analysis for $E[S]$.

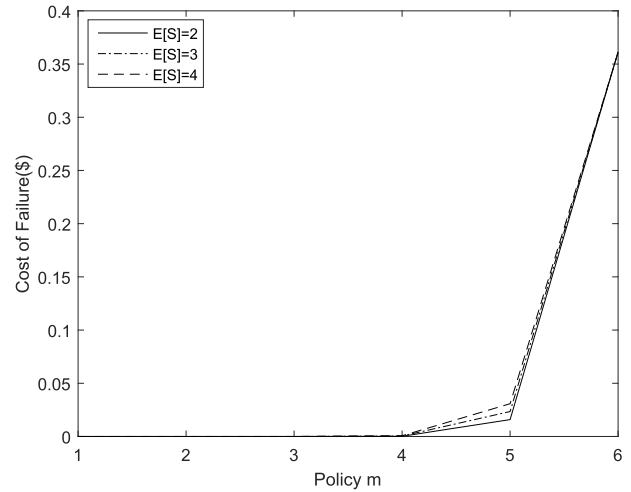
Explanation: As traffic intensity increases, server utilization also increases as does average queue length, server degradation

Fig. 7. Case 1: sensitivity analysis for $E[R]$.

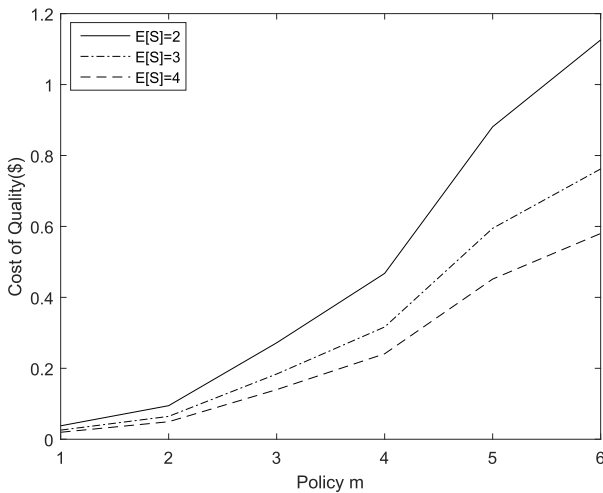
(a)



(a)



(b)



(b)

Fig. 8. Case 1: sensitivity analysis for $E[S]$. (a) Cost of holding. (b) Cost of quality.Fig. 9. Case 1: sensitivity analysis for $E[S]$. (a) Cost of repair. (b) Cost of failure.

Observation 2: As mean service time decreases, total cost increases. This effect is more pronounced for higher values of m (Fig. 6).

Explanation:

Cost of Holding: To hold traffic intensity constant as mean service time decreases, arrival rates must increase. This implies more arrivals during server repair, which causes average queue length to increase for fixed m . Thus, holding costs increase [Fig. 8(a)].

Cost of Quality: As mean service time decreases while holding traffic intensity constant, the number of services completed will increase for fixed m . Since quality cost is summed over services completed, quality costs will increase [Fig. 8(b)].

Cost of Repair: Changes in mean service time do not impact rates of repair, thus repair cost is constant for fixed m [Fig. 9(a)].

Cost of Catastrophic Failure: Shorter mean service time decreases the cost of failure for large m . This happens because

rate, and system output rate. An inspection of the objective function shows that all costs will, therefore, increase (Fig. 5).

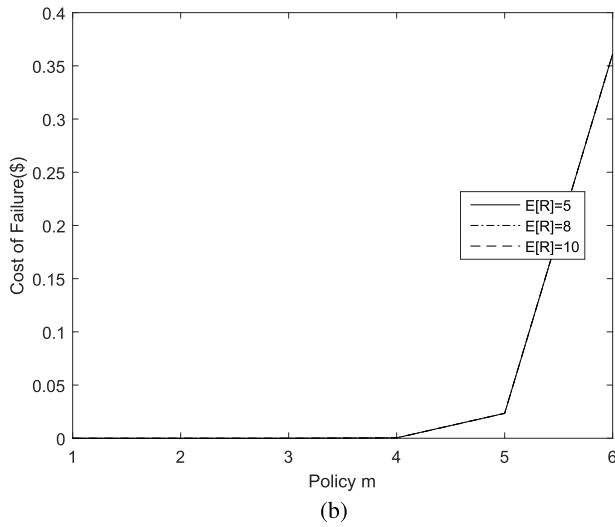
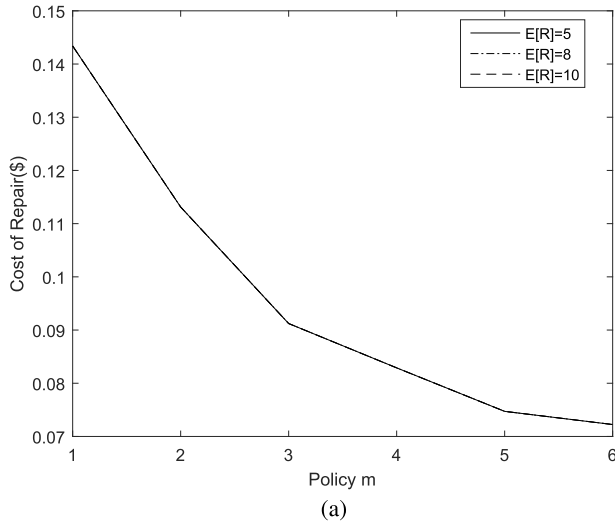


Fig. 10. Case 1: sensitivity analysis for $E[R]$. (a) Cost of repair. (b) Cost of failure.

the probability of unexpected failure decreases for shorter service times. However, the magnitudes of these cost increments are small relative to cost increases discussed about [Fig. 9(b)].

Observation 3: As mean repair time increases, total cost increases. This effect is less pronounced for higher values of m (Fig. 7).

Explanation: Changes in mean repair time do not impact quality, repair or failure (Figs. 10 and 11). However, longer repair times increase queues and holding cost [Fig. 11(b)]. Thus, mean repair time and total cost increase together.

V. CONCLUSION

This paper develops the queuing performance aspects of a service system with a single-degrading server that can be returned to a good-as-new state through a maintenance action. Results are presented for modeling the server degradation and repair processes, determining system stability, computing steady-state probabilities and long-term average costs, and identifying the optimal degradation state at which to initiate the server repair process. Costs in the objective function

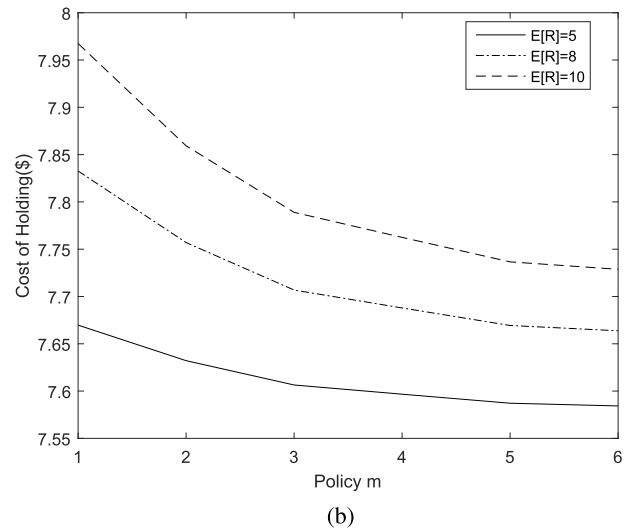
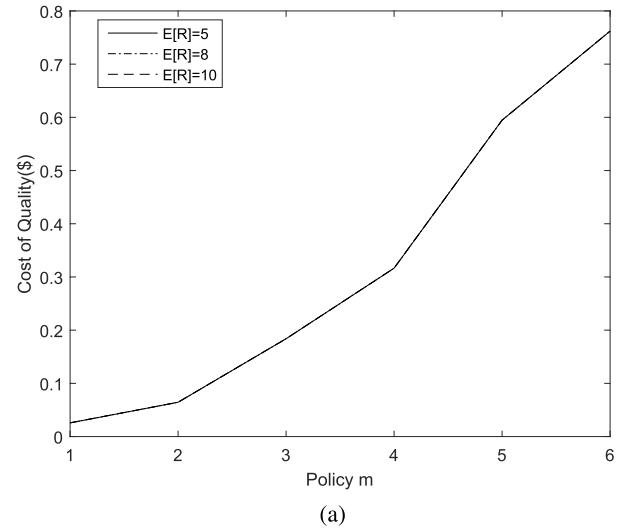


Fig. 11. Case 1: sensitivity analysis for $E[R]$. (a) Cost of quality. (b) Cost of holding.

include quality cost (which is assumed to increase with server degradation), maintenance cost, catastrophic failure cost, and inventory holding costs (or customer waiting costs). Service times are assumed constant and repair costs are stochastic. Numerical experiments are performed for model validation and sensitivity analysis using real degradation data. The model is easily applied to real systems through the collection of data on server and quality degradation, maintenance and service time requirements, and costs (or priorities) on quality, holding, catastrophic failure, and repair.

Future research includes extensions for stochastic service times, multiserver degradation, server degradation in queuing networks, and multidimensional policies that consider both server degradation and prevailing system workload.

APPENDIX DERIVATION OF c_k

As discussed in Section III, we can compute the probability that k arrivals occur during time Z for an arbitrary random

variable Z with CDF $H(t) = P\{Z \leq t\}$ as

$$\int_0^\infty e^{-\lambda t} \frac{(\lambda t)^k}{k!} dH(t)$$

for all $k \geq 0$.

A. Case 1: Constant Repair, Constant Service

For constant repair times, c_k is derived as follows:

$$\begin{aligned} c_k &= \int_0^\infty e^{-\lambda(s+(r-U)^+)} \frac{(\lambda(s+(r-U)^+))^k}{k!} \lambda e^{-\lambda u} du \\ &= \int_0^r e^{-\lambda(s+r-u)} \frac{(\lambda(s+r-u))^k}{k!} \lambda e^{-\lambda u} du \\ &\quad + \int_r^\infty e^{-\lambda s} \frac{(\lambda s)^k}{k!} \lambda e^{-\lambda u} du \\ &= e^{-\lambda(s+r)} \frac{\lambda^{k+1}}{k!} \int_0^r (s+r-u)^k du \\ &\quad + e^{-\lambda s} \frac{\lambda^{k+1} s^k}{k!} \int_r^\infty e^{-\lambda u} du \\ &= e^{-\lambda(s+r)} \frac{\lambda^{k+1}}{k!} \left(\frac{(s+r-u)^{k+1} (-1)}{k+1} \Big|_0^r \right) \\ &\quad + e^{-\lambda s} \frac{\lambda^{k+1} s^k}{k!} \left(\frac{e^{-\lambda u}}{-\lambda} \Big|_r^\infty \right) \\ &= e^{-\lambda(s+r)} \frac{\lambda^{k+1}}{k!} \left(\frac{(s+r)^{k+1} - s^{k+1}}{k+1} \right) \\ &\quad + e^{-\lambda s} \frac{\lambda^{k+1} s^k}{k!} \left(\frac{e^{-\lambda r}}{\lambda} \right) \\ &= \frac{e^{-\lambda(s+r)} \lambda^k}{k!} \left(\frac{\lambda}{k+1} ((s+r)^{k+1} - s^{k+1}) + s^k \right). \end{aligned}$$

B. Case 2: Weibull Repair, Constant Service

For Weibull repair times, c_k is derived as follows. α is the shape parameter and β is the rate parameter

$$\begin{aligned} 3c_k &= \int_0^\infty \int_0^r e^{-\lambda(s+r-u)} \frac{(\lambda(s+r-u))^k}{k!} \\ &\quad \alpha \beta (r\beta)^{\alpha-1} e^{-(r\beta)^\alpha} \lambda e^{-\lambda u} du dr \\ &\quad + \int_0^\infty \int_r^\infty e^{-\lambda s} \frac{(\lambda s)^k}{k!} \alpha \beta (r\beta)^{\alpha-1} e^{-(r\beta)^\alpha} \lambda e^{-\lambda u} du dr \\ c_k &= c'_k + c''_k \\ c'_k &= \int_0^\infty \int_0^r e^{-\lambda(s+r-u)} \frac{(\lambda(s+r-u))^k}{k!} \\ &\quad \alpha \beta (r\beta)^{\alpha-1} e^{-(r\beta)^\alpha} \lambda e^{-\lambda u} du dr \\ &= \int_0^\infty \int_0^r e^{-\lambda s} e^{-(r\beta)^\alpha - \lambda r} \frac{(\lambda(s+r-u))^k}{k!} \\ &\quad \alpha \beta (r\beta)^{\alpha-1} \lambda du dr \\ &= \int_0^\infty e^{-\lambda s} e^{-(r\beta)^\alpha - \lambda r} \left(\frac{(s+r-u)^{k+1} (-1)}{k+1} \Big|_0^r \right) \frac{\lambda^{k+1}}{k!} \\ &\quad \alpha \beta (r\beta)^{\alpha-1} dr \\ &= \int_0^\infty e^{-\lambda s} e^{-(r\beta)^\alpha - \lambda r} ((s+r)^{k+1} - s^{k+1}) \frac{\lambda^{k+1}}{(k+1)!} \\ &\quad \alpha \beta (r\beta)^{\alpha-1} dr \\ c''_k &= \int_0^\infty \int_r^\infty e^{-\lambda s} \frac{(\lambda s)^k}{k!} \end{aligned}$$

$$\begin{aligned} &\times \alpha \beta (r\beta)^{\alpha-1} e^{-(r\beta)^\alpha} \lambda e^{-\lambda u} du dr \\ &= \int_0^\infty e^{-\lambda s} \frac{(\lambda s)^k}{k!} \alpha \beta (r\beta)^{\alpha-1} e^{-(r\beta)^\alpha} \lambda \left(\frac{e^{-\lambda u}}{-\lambda} \Big|_r^\infty \right) dr \\ &= \int_0^\infty e^{-\lambda s} \frac{(\lambda s)^k}{k!} \alpha \beta (r\beta)^{\alpha-1} e^{-(r\beta)^\alpha} e^{-\lambda r} dr \\ &= \int_0^\infty e^{-\lambda s} e^{-(r\beta)^\alpha - \lambda r} \frac{(\lambda s)^k}{k!} \alpha \beta (r\beta)^{\alpha-1} dr \\ c_k &= \int_0^\infty e^{-\lambda s} e^{-(r\beta)^\alpha - \lambda r} ((s+r)^{k+1} - s^{k+1}) \frac{\lambda^{k+1}}{(k+1)!} \\ &\quad \alpha \beta (r\beta)^{\alpha-1} dr \\ &\quad + \int_0^\infty e^{-\lambda s} e^{-(r\beta)^\alpha - \lambda r} \frac{(\lambda s)^k}{k!} \alpha \beta (r\beta)^{\alpha-1} dr. \end{aligned}$$

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